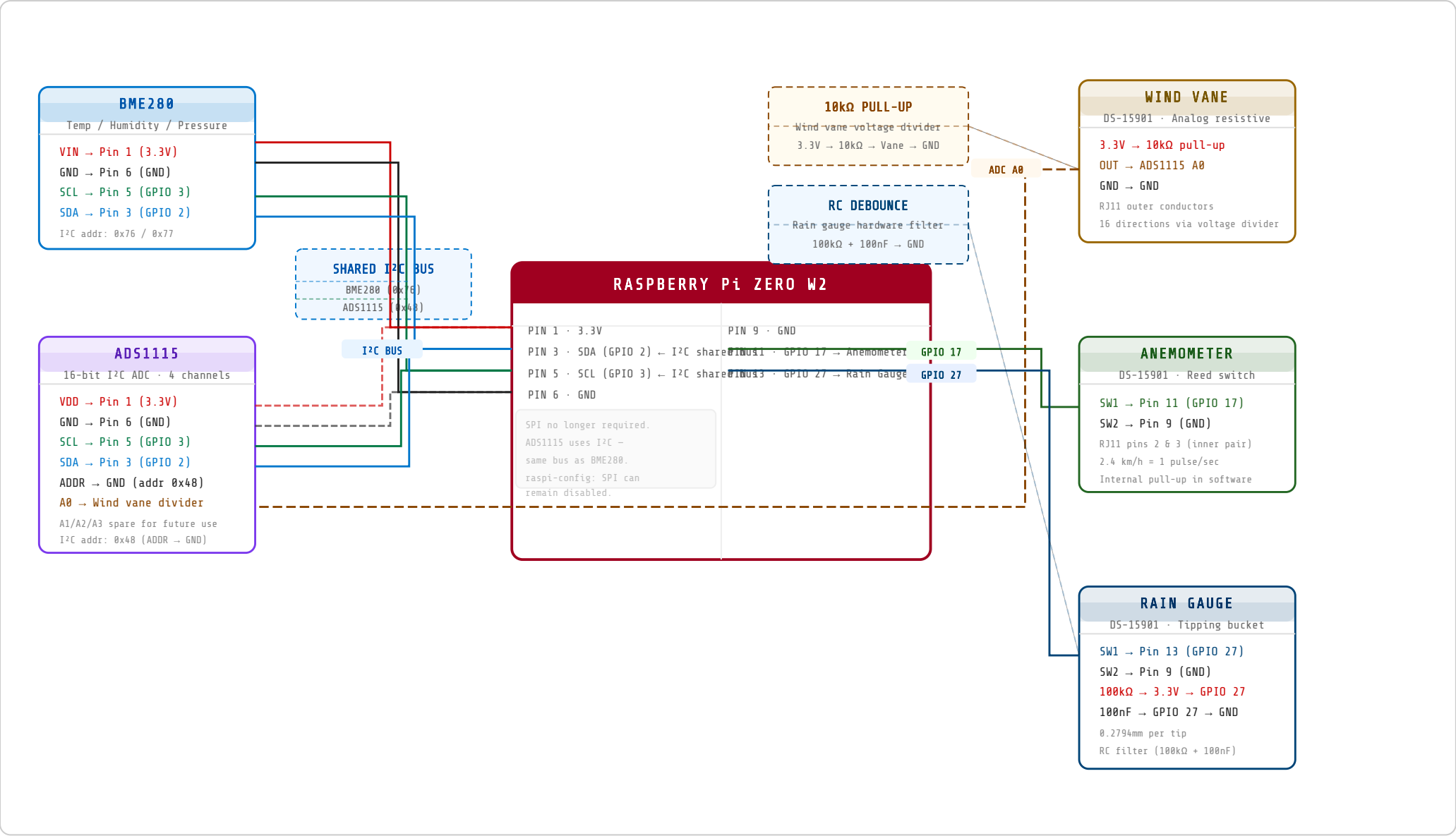


WEATHER STATION – CONNECTION DIAGRAM

Raspberry Pi Zero W2 · BME280 (I²C) · ADS1115 ADC (I²C) · DS-15901 Anemometer / Wind Vane / Rain Gauge (GPIO)

⚠ Updated diagram – ADS1115 replaces MCP3008. Both BME280 and ADS1115 now share the same I²C bus (SDA/SCL). SPI bus and CE0 are no longer used. Only 4 wires needed for both I²C devices combined.



3.3V Power

GND

I²C SDA (GPIO 2)

I²C SCL (GPIO 3)

Anemometer (GPIO 17)

Rain Gauge (GPIO 27)

Wind Vane → ADC A0

Shared rail (dashed)

BME280 → PI ZERO W2

BME280	PI PIN #	SIGNAL
VIN	1	3.3V
GND	6	GND
SCL	5	GPIO 3 / SCL
SDA	3	GPIO 2 / SDA

I²C address: 0x76 (default) or 0x77 · SDO pin sets address

ANEMOMETER → PI ZERO W2

ANEMOMETER	PI PIN #	SIGNAL
Reed SW wire 1	11	GPIO 17
Reed SW wire 2	9	GND

RJ11 pins 2 & 3 · 2.4 km/h = 1 pulse/sec · pull-up in software

ADS1115 → PI ZERO W2

ADS1115	PI PIN #	SIGNAL
VDD	1	3.3V
GND	6	GND
SCL	5	GPIO 3 / SCL
SDA	3	GPIO 2 / SDA
ADDR	6	GND → addr 0x48
A0	—	Wind vane divider

Shares SDA/SCL with BME280 · A1-A3 spare · no SPI needed

WIND VANE → ADS1115

CONNECTION	TO	NOTES
3.3V	10kΩ top	Pull-up supply
Junction	A0 + Vane+	Divider mid-point
Vane-	GND	RJ11 outer pair

Voltage → resistance → 16 directions (0°-337.5°)

RAIN GAUGE → PI ZERO W2

RAIN GAUGE	PI PIN #	SIGNAL
Reed SW wire 1	13	GPIO 27
Reed SW wire 2	9	GND
100kΩ pull-up	1	3.3V → GPIO 27
100nF cap	9	GPIO 27 → GND

RJ11 centre conductors · 0.2794mm per tip · 100nF recommended